



Universidade de Brasília
Faculdade de Tecnologia
Departamento de Engenharia Mecânica

**Gated Data-Space Inversion for
Reservoir Prediction**

Thiago Tomás de Paula

DISSERTAÇÃO DE MESTRADO
PROGRAMA DE PÓS-GRADUAÇÃO EM CIÊNCIAS MECÂNICAS

Brasília
2026

Universidade de Brasília
Faculdade de Tecnologia
Departamento de Engenharia Mecânica

Agregação de Modelos Adaptativa
Aplicada a Otimização de Reservatórios

Thiago Tomás de Paula

Dissertação de Mestrado submetida ao Programa de Pós-Graduação em Ciências Mecânicas como requisito parcial para obtenção do grau de Mestre.

Orientador: Prof. Dr. Eugênio Libório Feitosa Fortaleza

Brasília
2026

FICHA CATALOGRÁFICA

P324g Paula, Thiago Tomás de.
Gated Data-Space Inversion for Reservoir Prediction / Thiago Tomás de Paula; orientador Eugênio Libório Feitosa Fortaleza. -- Brasília, 2026.
25 p.

Dissertação de Mestrado (Programa de Pós-Graduação em Ciências Mecânicas) -- Universidade de Brasília, 2026.

1. . 2. . 3. . I. Fortaleza, Eugênio Libório Feitosa, orient. II. Título.

Universidade de Brasília
Faculdade de Tecnologia
Departamento de Engenharia Mecânica

Gated Data-Space Inversion for Reservoir Prediction

Thiago Tomás de Paula

Dissertação de Mestrado submetida ao Programa de Pós-Graduação em Ciências Mecânicas como requisito parcial para obtenção do grau de Mestre.

Trabalho aprovado. Brasília, 15 de julho de 2026:

Prof. Dr. Eugênio Libório Feitosa Fortaleza
UnB/FT/ENM
Orientador

Examinador interno

Examinador interno

Abstract

Guiding questions

- What is the reason for writing the thesis?
- What are the current approaches and gaps in the literature?
- What are your research question(s) and aims?
- Which methodology have you used?
- What are the main findings?
- What are the main conclusions and implications?

Accurate prediction of reservoir performance under uncertainty is essential for decision-making in geological reservoir management. Data-Space Inversion (DSI) ([Sun; Durlowsky, 2017](#)) provides an attractive alternative to traditional history matching by updating ensemble forecasts directly in data space, thereby avoiding repeated subsurface simulations. Recently, [Lima, Emerick, and Ortiz \(2020\)](#) introduced DSI-ESMDA, an extension of DSI that employs the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) to replace the Randomized Maximum Likelihood minimizations of the original formulation with an iterative Kalman-type update, achieving significant computational efficiency while maintaining competitive posterior forecasts. Despite these advantages, DSI-ESMDA, like other ensemble smoothers, remains susceptible to excessive variance loss, which can lead to underestimation of posterior uncertainty. Methods such as covariance localization help to ameliorate this issue, but variance underestimation is still observed in practical applications. One contributing factor to this progressive loss of variance is that small production mismatches, often dominated by measurement noise or sampling effects, may trigger repeated micro-updates that reduce ensemble diversity without improving predictive skill. This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. In contrast to covariance localization, which acts on second-order statistics, innovation gating acts directly on the update term, selectively suppressing corrections associated with sufficiently-matched observations while preserving the full impact of significant mismatches. Match quality is thresholded by a gating parameter during innovation normalization. The gating methodology is inspired by adaptive filtering developed to mitigate actuator wear in closed-loop control systems ([Fortaleza et al., 2022](#)) and is conceptually related to smooth shrinkage and noise-gating strategies in signal processing ([Donoho; Johnstone, 1994; Donoho, 1995](#)). Although the present work focuses on DSI, the proposed formulation is general and can be applied to standard ensemble-based model-space

inversion methods, including ES-MDA and related iterative smoothers. The approach is evaluated on multiple synthetic reservoirs and one real field application using production rates from producers and injectors. Results show that innovation-gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match. Furthermore, for moderate gating parameters, posterior forecasts remain close to those obtained with standard DSI-ESMDA, indicating that the proposed method acts as a stabilizing extension rather than a replacement.

Keywords: ; ; .

Resumo

Palavras-chave: ; ; .

Contents

1	Introduction	9
1.1	Work overview	9
1.2	Background context	10
1.3	Research problem	12
1.4	Research questions	12
1.5	Significance	12
1.6	Methodology overview	13
1.7	Dissertation outline	13
2	Literature Review	15
2.1	Chapter outline	15
2.2	Chapter summary	16
3	Theory framework	17
3.1	Chapter outline	17
3.2	Elements of reservoir engineering	17
3.2.1	Reservoir formation	17
3.2.2	Reservoir simulation	17
3.3	Kalman Filtering	17
3.4	Data-Space Inversion	18
3.5	Chapter summary	18
4	Methods	19
4.1	Chapter outline	19
4.2	Reservoir case studies	19
4.2.1	Egg ensemble	19
4.2.2	Olympus ensemble	19
4.2.3	Real field	19
4.3	Chapter summary	19
5	Results	20
5.1	Chapter outline	20
5.2	Egg ensemble	20
5.3	Olympus ensemble	20
5.4	Real field	20
5.5	Chapter summary	20
6	Discussion	21
6.1	Chapter outline	21

6.2	Summary of Key Findings	22
6.3	Interpretation of Findings	22
6.4	Evaluation of Existing Theories and Models	22
6.5	Limitations and Future Research	22
6.6	Chapter summary	22
7	Conclusion	23
7.1	Summary of Findings	23
7.2	Contribution to Knowledge	23
7.3	Practical Implications	23
7.4	Limitations	23
7.5	Future Research	24
7.6	Closing Statement	24
	References	25

1 Introduction

Guiding questions

- What is your original contribution? Why should the examiner care about your research?
- What is the thesis problem statement?
- What do you (not) hope to achieve?
- What are the research questions and hypotheses?
- What are your epistemological and ontological positions?
- What is your contribution to the field?
- How is the thesis laid out?

PhD introduction at a glance:

Element	Purpose	Position	Typical Length
Background Context	Set the scene for your research	Opening section	500-1,000 words
Research Problem	Define the specific gap or issue	After context	300-500 words
Research Questions	State your precise questions	After problem	200-400 words
Significance	Explain why this research matters	After questions	300-500 words
Methodology Overview	Brief summary of approach	After significance	200-400 words
Thesis Outline	Chapter-by-chapter roadmap	Final section	300-500 words

Once you have finished your thesis, ask yourself the following questions:

- Does the first line of the introduction discuss the problem that your thesis is addressing and the contribution that it is making?
- Does the introduction provide an overview of the thesis and end with a brief discussion on the content of each chapter?
- Does the introduction make a case for the research?
- Have the research questions/problems/hypotheses been clearly outlined (preferably early on)?

1.1 Work overview

- Introduction to the introduction: a short version (of only a few paragraphs) of the thesis' aims, research questions, contribution, objectives and findings.

The economic success of oil and gas fields is increasingly contingent on the quality of reservoir management decisions, which are made under uncertainty and based on limited data. Kalman-based ensemble methods are widely used to quantify uncertainty in reservoir

forecasts by adjusting reservoir models to match production data, but require vast computational resources to perform repeated reservoir simulations. In an effort to reduce the computational burden, Data-Space Inversion (DSI) (Sun; Durlofsky, 2017) was introduced as an alternative to traditional history matching by updating ensemble forecasts directly in data space, thereby avoiding repeated subsurface simulations.

Lima, Emerick, and Ortiz (2020) introduced DSI-ESMDA, an extension of DSI that employs the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) to replace the Randomized Maximum Likelihood minimizations of the original formulation with an iterative Kalman-type update, achieving significant computational efficiency while maintaining competitive posterior forecasts. However, DSI-ESMDA, like related ensemble smoothers, remains susceptible to excessive variance loss, which can lead to underestimation of posterior uncertainty.

This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. Findings on both synthetic and real reservoirs show that Gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match.

Gated DSI-ESMDA acts as a stabilizing extension rather than a replacement, and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The power of this contribution is that it provides a simple, yet effective means of improving the robustness of ensemble-based inversion methods without requiring additional simulations or complex modifications to existing workflows.

1.2 Background context

- State the overarching topic and aims of the thesis in more detail
- Provide a brief review of the literature related to the topic (this will be very brief if you have a separate literature review chapter)
- Define the terms and scope of the topic

The economic viability of oil and gas operations depends critically on sound reservoir management, where decisions must be made under significant uncertainty and with inherently limited subsurface data. Accurate characterization of this uncertainty is essential: overconfident forecasts can lead to poor investment decisions, while excessively uncertain ones impede operational planning. History matching — the process of calibrating reservoir models to observed production data — has long been central to reducing forecast uncertainty,

and ensemble-based methods have become the dominant paradigm for performing this calibration at scale.

Kalman-based ensemble methods, such as the Ensemble Smoother with Multiple Data Assimilation (ES-MDA), address this problem by maintaining an ensemble of plausible reservoir models and updating them iteratively as production data are assimilated. By propagating uncertainty through an ensemble rather than relying on a single deterministic model, these methods provide a principled framework for quantifying forecast uncertainty. However, they carry a substantial computational cost: each update cycle requires running full reservoir simulations for every ensemble member, making large-scale uncertainty quantification expensive and, in practice, often infeasible.

Data-Space Inversion (DSI), introduced by Sun and Durlofsky (2017), was proposed as a computationally efficient alternative to traditional model-space history matching. Rather than updating subsurface model parameters and re-running simulations, DSI operates directly on the ensemble of simulated production forecasts — working entirely in data space. This eliminates the need for repeated reservoir simulations during the assimilation process, dramatically reducing the computational burden while still yielding calibrated posterior forecast distributions.

Building on this foundation, Lima, Emerick, and Ortiz (2020) developed DSI-ESMDA, which replaces the Randomized Maximum Likelihood minimizations of the original DSI formulation with an iterative ES-MDA-style Kalman update. This modification brings further gains in computational efficiency and produces competitive posterior forecasts, establishing DSI-ESMDA as a practical and attractive framework for reservoir uncertainty quantification.

Despite these advances, DSI-ESMDA inherits a well-known vulnerability of ensemble smoother methods: excessive variance loss during data assimilation. When observations strongly constrain the ensemble, the spread of forecasts can collapse beyond what is physically justified, leading to an underestimation of posterior uncertainty. This phenomenon is particularly problematic in reservoir forecasting, where robust uncertainty envelopes are essential for risk-aware decision-making. The issue is compounded in data-space methods, where the ensemble is not replenished through new simulations and the initial ensemble spread is the only source of diversity throughout the assimilation process.

Mitigating variance collapse without sacrificing data match quality or introducing additional computational overhead remains an open challenge. Existing remedies — such as inflation schemes and localization — typically require careful tuning and may not transfer cleanly to the data-space setting. This motivates the development of a principled, lightweight mechanism that can stabilize the Kalman update in DSI-ESMDA, preserving ensemble diversity and producing more reliable uncertainty quantification across a range of reservoir scenarios.

1.3 Research problem

- Critically evaluate the current state of the literature on that topic and identify your gap
- Outline why the research is important and the contribution that it makes
- Outline your epistemological and ontological position

Literature on DSI and DSI-ESMDA take different approaches to mitigating variance loss, but they are either computationally expensive or require careful tuning of hyperparameters. Ideally, a solution would be simple to implement, require minimal tuning, and be computationally efficient. This work proposes introducing a straightforward gating mechanism on DSI-ESMDA's innovation vector, which selectively suppresses the influence of well-matched observations while preserving the impact of significant mismatches. Gated DSI-ESMDA thus modulates the posterior ensemble variance with a single parameter, the intensity of the gating function.

1.4 Research questions

- Clearly outline the research questions and problem(s) you seek to address

This work addresses the following research questions:

1. Can a gating mechanism be introduced into DSI-ESMDA to selectively modulate the influence of observations based on their match quality?
2. Does the proposed innovation gating reduce variance loss during assimilation and produce more robust uncertainty envelopes in the prediction phase without degrading the data match?
3. How does the performance of Gated DSI-ESMDA compare to standard DSI-ESMDA across synthetic and real reservoir case studies?

1.5 Significance

- Restate the importance of the research in more detail

This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. In contrast to covariance localization, which acts on second-order statistics, innovation gating acts directly on the update term, selectively suppressing corrections associated with sufficiently-matched observations while preserving the full impact of significant mismatches. Match quality is thresholded by a gating parameter

during innovation normalization. The gating methodology is inspired by adaptive filtering developed to mitigate actuator wear in closed-loop control systems (Fortaleza *et al.*, 2022) and is conceptually related to smooth shrinkage and noise-gating strategies in signal processing (Donoho; Johnstone, 1994; Donoho, 1995). Although the present work focuses on DSI, the proposed formulation is general and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The proposed method inherits the computational efficiency of DSI-ESMDA while providing a robust, easily tunable mechanism for modulating posterior distribution variance, setting it as a practical and effective tool for uncertainty quantification in reservoir forecasting.

1.6 Methodology overview

- State the hypotheses (if you are using any)
- Detail the most important concepts and variables
- Briefly describe your methodology
- Discuss the main findings

The approach is evaluated against the original DSI-ESMDA on two synthetic reservoirs and one real field application using production rates from producers and injectors. The synthetic reservoirs are the Egg and Olympus ensembles, which are widely used in the literature to benchmark history matching and uncertainty quantification methods, while the real field application is a mature oil field in the Brazilian pre-salt. Posterior and prior ensembles are compared across several metrics, including observation coverage and mean squared error, to assess the impact of innovation gating on variance loss and uncertainty quantification. Results show that innovation-gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match. Furthermore, for moderate gating parameters, posterior forecasts remain close to those obtained with standard DSI-ESMDA, indicating that the proposed method acts as a stabilizing extension rather than a replacement.

1.7 Dissertation outline

- Discuss the layout of the thesis

Chapter 2 presents a review of the literature on reservoir uncertainty quantification, focusing on ensemble-based methods and data-space inversion. Chapter 3 introduces the theoretical framework underpinning the proposed methodology, from the relevant elements of reservoir engineering as a whole to the mathematical formulation of DSI-ESMDA and the innovation gating mechanism. Chapter 4 details the methods used to evaluate the pro-

posed approach, including descriptions of the synthetic and real reservoir case studies, data collection, and analysis techniques. [Chapter 5](#) presents the results of applying innovation-gated DSI-ESMDA to the case studies, including quantitative and qualitative assessments of performance relative to standard DSI-ESMDA. [Chapter 6](#) discusses the implications of the findings and their relationship to the existing literature. [Chapter 7](#) summarizes the main contributions and provides final remarks on the research and future work.

2 Literature Review

2.1 Chapter outline

Aims and structure

According to Hart (2018), the job of a literature review is to show five things (if you're using our free PhD Writing Template, you may recognise these):

- What has been written on your topic
- Who the key authors are and what the key works are
- The main theories and hypotheses
- The main themes that exist in the literature
- Gaps and weaknesses that your study will then help fill

A literature review has three objectives:

- Summarise what has already been discussed in your field, both to demonstrate that you understand your field and to show how your study relates to it.
- Highlight gaps, problems or shortcomings in existing research to show the original contribution that your thesis makes.
- Identify important studies, theories, methods or theoretical frameworks that can be applied in your research.

There are nine steps involved in conducting a literature review:

- Pick a broad topic
- Find the way in
- Who's saying what and when
- Take notes
- Narrow down the field
- Narrow down the sources
- Snowball
- Think about questions that haven't been asked
- Write early, write quickly and write relevantly

The DSI method introduced by Sun and Durlofsky [419] employs PCA to reparameterize the predicted data from the prior ensemble into a lower-dimensional space and uses RML to sample from the posterior distribution of predicted data. The authors applied a data transformation prior to PCA to enhance linearity. Sun et al. [421] further extended the DSI method by introducing a more generalized data transformation procedure, and Sun and Durlofsky [420] applied the method to a geological carbon storage problem. Earlier, Scheidt et al. [381] proposed a similar

concept in a method called prediction-focused analysis (PFA), which involves projecting prior predictions into a low-dimensional space and using kernel smoothing to estimate the joint distribution of historical and forecasted data. However, PFA seems to be limited to problems with a small number of data points, as it requires projection into very few dimensions (two or three in the examples presented). Satija and Caers [379] improved the PFA method by incorporating canonical functional component analysis to enhance linearity in the projected data, and Satija and Caers [380] applied this approach to a reservoir problem, finding that it provided uncertainty estimates for production forecasts in reasonable agreement with rejection sampling. Lima et al. [263] combined ES-MDA and DSI, resulting in a more robust and efficient DSI implementation. This method has been further enhanced through the use of various deep learning techniques, leading to successful applications in naturally fractured reservoirs [218] and geological carbon storage [469, 217] (Emerick, 2025).

2.2 Chapter summary

Key takeaways

3 Theory framework

3.1 Chapter outline

Aims and structure

The theory framework is the scaffolding upon which your thesis is built. When you're done writing your theory framework chapter or section, your reader should be able to answer these questions:

- What theoretical concepts are used in the research? What hypotheses, if any, are you using?
- Why have you chosen this theory?
- What are the implications of using this theory?
- How does the theory relate to the existing literature, your problem statement and your epistemological and ontological positions?
- How has this theory has been applied by others in similar contexts? What can you learn from them and how do you differ?
- How do you apply the theory and measure the concepts (with reference to the literature review/problem statement)?
- What is the relationship between the various elements and concepts within the model? Can you depict this visually?

That means that a theory framework can take different forms:

- It can state the theoretical assumptions underpinning the study.
- It can connect the empirical data to existing knowledge.
- It can allow you to come up with propositions, concepts or hypotheses that you can use to answer 'how' and 'why' questions.

3.2 Elements of reservoir engineering

3.2.1 Reservoir formation

3.2.2 Reservoir simulation

3.3 Kalman Filtering

3.4 Data-Space Inversion

3.5 Chapter summary

Key takeaways

4 Methods

The job of a methods chapter is:

- To summarise, explain and recount how you answered your research questions and to explain how this relates to the methods used by other scholars in similar contexts and similar studies
- To discuss - in detail - the techniques you used to collect the data used to answer your research questions
- To discuss why the techniques are relevant to the study's aims and objectives
- To explain how you used them

Your reader should be able to answer the following questions when they're done reading it:

- What did you do to achieve the research aims?
- Why did you choose this particular approach over others?
- How does it relate to your epistemological and ontological positions?
- What tools did you use to collect data and why? What are the implications?
- When did you collect data, and from whom?
- What tools have you used to analyse the data and why? What are the implications? Are there ethical considerations to take into account?

4.1 Chapter outline

Aims and structure

4.2 Reservoir case studies

4.2.1 Egg ensemble

4.2.2 Olympus ensemble

4.2.3 Real field

4.3 Chapter summary

Key takeaways

5 Results

When your reader is finished with this section, they should be able to answer the following questions:

- What are the results of your investigations?
- How do the findings relate to previous studies?
- Was there anything surprising or that didn't work out as planned?
- Are there any themes or categories that emerge from the data?
- Have you explained to the reader why you have reached particular conclusions?
- Have you explained the results?

5.1 Chapter outline

Aims and structure

5.2 Egg ensemble

5.3 Olympus ensemble

5.4 Real field

5.5 Chapter summary

Key takeaways

6 Discussion

6.1 Chapter outline

Aims and structure

You are providing sufficient detail that others can draw their own inferences and construct their own explanations. Think of it as presenting the case for a jury. That means that an empirical discussion should:

- Tell the reader how the data was collected, with reference to the methods chapter/section
- Tell them how they can access it if they wanted to replicate your study
- Discuss what the results look like (using visual aids, such as tables, diagrams, graphs and so on)
- Provide rich summaries of the findings
- Discuss the gaps in the findings and analysis
- Analyse the results
- Discuss the implications of your findings
- Discuss the limitations of the findings

When it comes to writing your discussion chapter, you can start by writing a few sentences that summarise the most important results. One danger when writing discussion sections is that they can be too wordy, offer too much interpretation and lack a clear structure. To avoid this, you should make sure that every element of your discussion section addresses one of the following questions:

- What are the relationships between observations? (The mud-map you developed earlier will help here.)
- Are there any trends and generalisations amongst the results? Are there any exceptions to these?
- What are the causes of, or mechanisms behind, the underlying patterns you have uncovered?
- Do your results agree or disagree with previous work?
- How do your findings relate to the theoretical framework you developed, if applicable?
- How do the findings relate to the hypotheses you developed, if applicable?
- What other explanations could there be for your results? This issue is more pertinent if you are engaging in theory creation/inductive reasoning.
- What do we now know as a result of your research that we didn't know before?

- What is the significance of these findings?
- Why should we care about the findings?

When your discussion chapter is finished, your reader should be able to answer the following questions:

- How do the findings relate to the theory and methods discussed previously?
- Why you have reached particular conclusions?
- How do your findings relate to the gaps in the literature you identified earlier?
- What implications do the findings have for the discipline and for existing understanding?
- How do the findings relate to your research questions/aims and objectives?

6.2 Summary of Key Findings

- Brief Restatement of Research Findings
- Comparison with Initial Hypotheses or Research Questions

6.3 Interpretation of Findings

- Contextualisation of Results
- Significance and Implications of the Findings

6.4 Evaluation of Existing Theories and Models

- How Your Findings Support or Challenge Previous Work
- Conceptual Contributions of Your Study

6.5 Limitations and Future Research

- Acknowledgment of Study Limitations
- Suggestions for Future Research

6.6 Chapter summary

Key takeaways

7 Conclusion

The job of the conclusion is to:

- Fully and clearly articulate the answer to your research questions
- Discuss how the research is related to your aims and objectives
- Explain the significance of the work
- Outline its shortcomings
- Suggest avenues for future research

It is not the place to introduce new ideas and concepts, or to present new findings. Your job is to reflect back on your original aims and intentions and discuss them in terms of your findings and new expertise.

By the time the reader has finished reading the conclusion, they should be able to answer the following questions:

- Have you briefly recapped the research questions and objectives?
- Have you provided a brief recount of the answer to those questions?
- Have you clearly discussed the significance and implications of those findings?
- Have you discussed the contribution that the study has made?
- Do the claims you are making align with the content of the results and discussion chapters?

7.1 Summary of Findings

Briefly restate key results

7.2 Contribution to Knowledge

State what is new and original

7.3 Practical Implications

How findings apply in practice

7.4 Limitations

Honest assessment of constraints

7.5 Future Research

Suggest next steps for the field

7.6 Closing Statement

Final reflection on significance

References

- DONOHU, D. De-noising by soft-thresholding. **IEEE Transactions on Information Theory**, v. 41, n. 3, p. 613–627, 1995. DOI [10.1109/18.382009](https://doi.org/10.1109/18.382009). Cit. on pp. 4 and 13.
- DONOHU, D. L.; JOHNSTONE, I. M. Ideal spatial adaptation by wavelet shrinkage. **Biometrika**, v. 81, n. 3, p. 425–455, 09 1994. ISSN 0006-3444. DOI [10.1093/biomet/81.3.425](https://doi.org/10.1093/biomet/81.3.425). Available at: <https://doi.org/10.1093/biomet/81.3.425>. Cit. on pp. 4 and 13.
- EMERICK, A. **Ensemble Data Assimilation Applied to Geological Reservoir Models**. [S.l.: s.n.], 2025. ISBN 978-65-88763-29-2. Cit. on p. 16.
- FORTALEZA, E. L.; GOMES, L. M.; de A. Limaverde Filho, J. O.; de Campos, M. C.; LONGHI, L. G. S.; de Lima, M. L.; TOGNETTI, E. S. Model-free adaptive filter to mitigate actuator wear. **ISA Transactions**, v. 129, p. 493–504, 2022. ISSN 0019-0578. DOI <https://doi.org/10.1016/j.isatra.2022.02.026>. Available at: <https://www.sciencedirect.com/science/article/pii/S0019057822000829>. Cit. on pp. 4 and 13.
- LIMA, M. M.; EMERICK, A. A.; ORTIZ, C. E. P. Data-space inversion with ensemble smoother. **Computational Geosciences**, v. 24, n. 3, p. 1179–1200, Jun 2020. ISSN 1573-1499. DOI [10.1007/s10596-020-09933-w](https://doi.org/10.1007/s10596-020-09933-w). Available at: <https://doi.org/10.1007/s10596-020-09933-w>. Cit. on pp. 4 and 10.
- SUN, W.; DURLOFSKY, L. J. A new data-space inversion procedure for efficient uncertainty quantification in subsurface flow problems. **Mathematical Geosciences**, v. 49, n. 6, p. 679–715, 2017. DOI [10.1007/s11004-016-9672-8](https://doi.org/10.1007/s11004-016-9672-8). Available at: <https://doi.org/10.1007/s11004-016-9672-8>. Cit. on pp. 4 and 10.



UnB